Frequency Transformations for Continuous-Time Systems

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Key Concepts

1) A frequency transformation is used to convert a prototype lowpass filter to a filter with the desired pass and stop band configuration.

2) This lesson covers frequency transformations for continuous-time systems because they are used in MATLAB's IIR filter design routines.

3) Frequency transformations remap the independent variable of the system transfer function to warp the frequency axis in a desired manner.

4) Lowpass to lowpass transformations

    a) Change the passband width of the lowpass prototype filter.

    b) Linear transformation of the frequency axis to scale the filter response to have the desired cutoff frequency.

5) Lowpass to highpass transformations

    a) Map the lowpass filter passband to high frequencies and lowpass filter stopband to low frequencies.

    b) Use an inverse transform to swap high- and low-frequency components of the frequency axis.

6) Lowpass to bandpass transformations

   a) Map the lowpass filter passband to a midband and the stop band to both high and low frequencies.

   b) Design parameters chosen to control passband center and bandwidth.

Video Screenshots

Frequency transformation —

convert prototype (usually lowpass) filter to desired filter

$s = f(\tilde{s})$ transformation relating s to $\tilde{s}$

$$H(\tilde{s}) = H_{LP}(s)\Big|_{s=f(\tilde{s})}$$

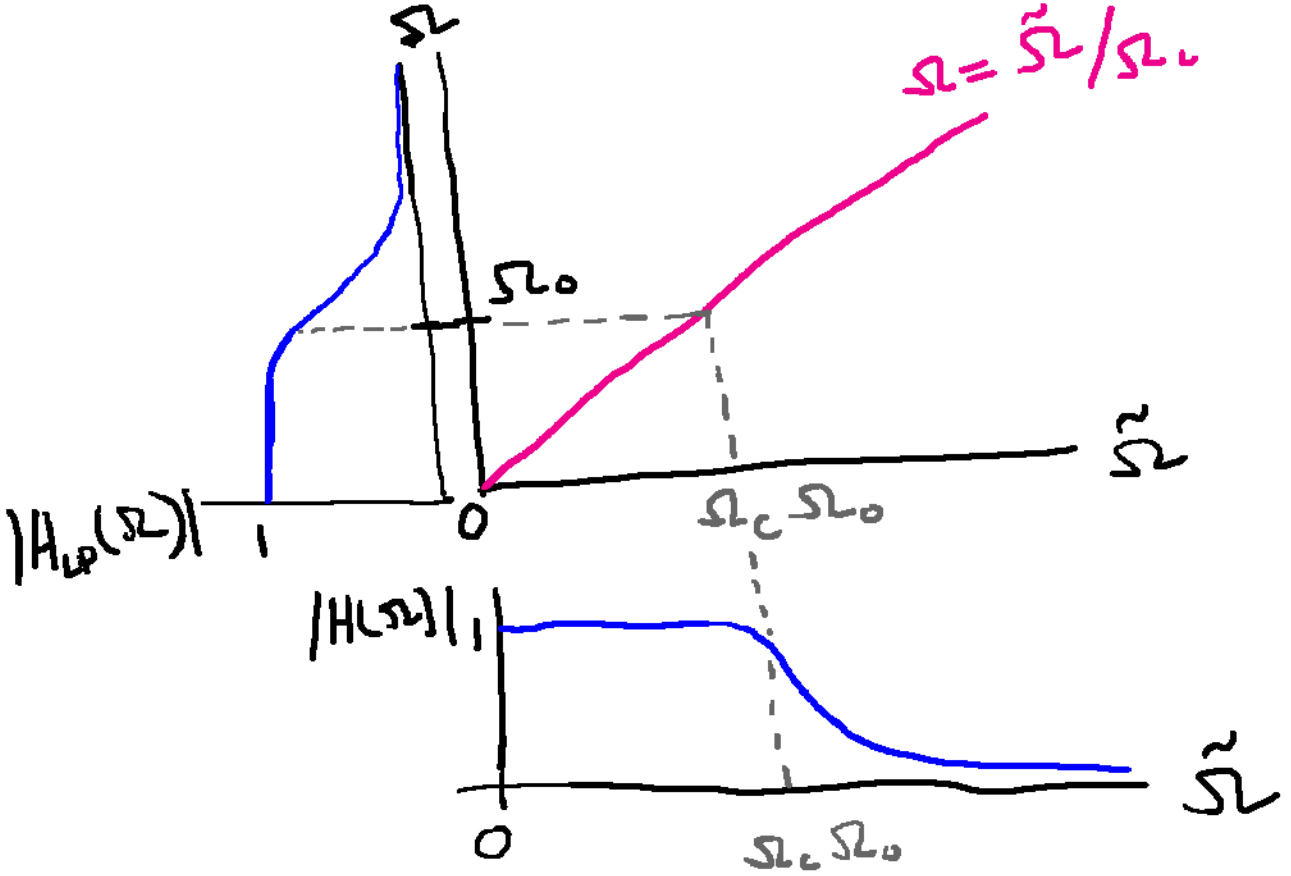
$$H(\tilde{\Omega}) = H_{LP}(\Omega)\Big|_{\Omega=f(\tilde{\Omega})}$$ "warp" frequency axis

Lowpass to lowpass
Lowpass to highpass
Low pass to bandpass

Lowpass to Lowpass

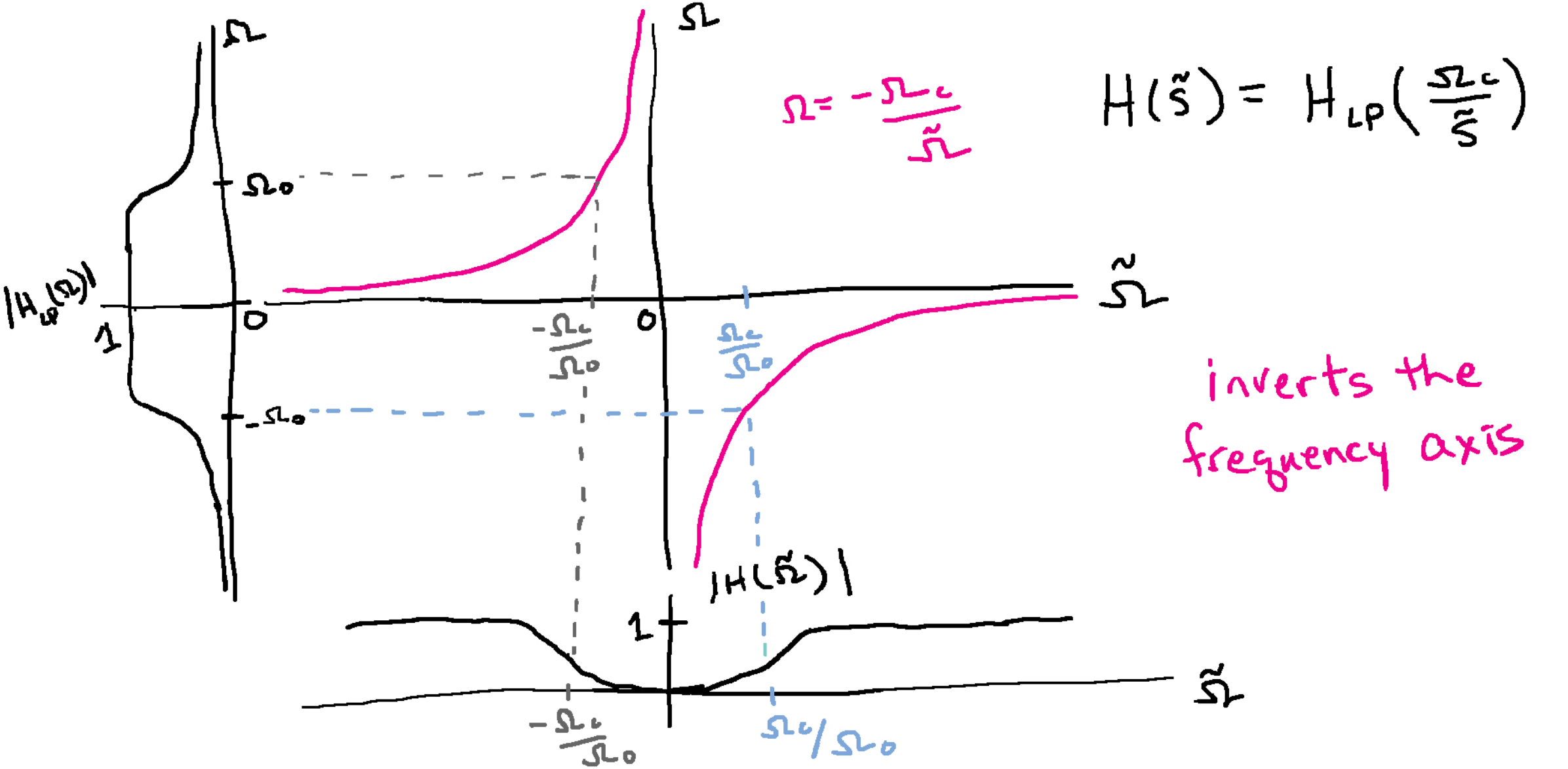
$s = \tilde{s}/\Omega_c \Rightarrow_{s=j\Omega} \Omega = \tilde{\Omega}/\Omega_c$

$H(\tilde{s}) = H_{LP}(\tilde{s}/\Omega_c)$



$\tilde{\Omega}$ axis is Ω axis stretched by Ω_c

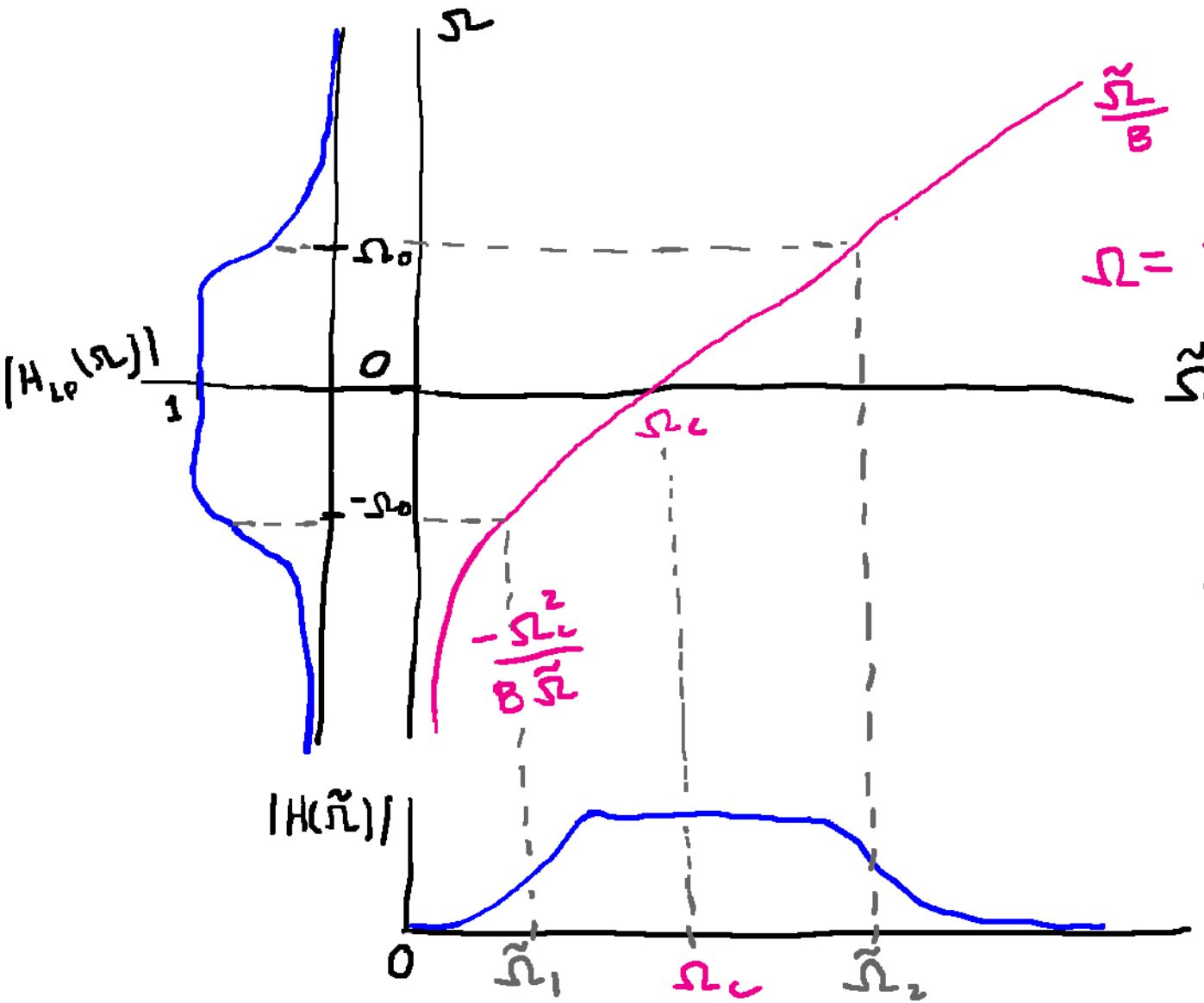
Lowpass to Highpass: $s = \frac{\Omega_c}{\tilde{s}} \Rightarrow_{s=j\Omega} \Omega = -\frac{\Omega_c}{\tilde{\Omega}}$



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Lowpass to Bandpass:

$$s = \frac{\tilde{s}^2 + \Omega_c^2}{B\tilde{s}} \Rightarrow_{s=j\Omega} \Omega = \frac{\tilde{\Omega}^2 - \Omega_c^2}{B\tilde{\Omega}}$$



$$H(\tilde{s}) = H_{LP}\left(\frac{\tilde{s}^2 + \Omega_c^2}{B\tilde{s}}\right)$$

$$\Omega = \frac{\tilde{\Omega}^2 - \Omega_c^2}{B\tilde{\Omega}}$$

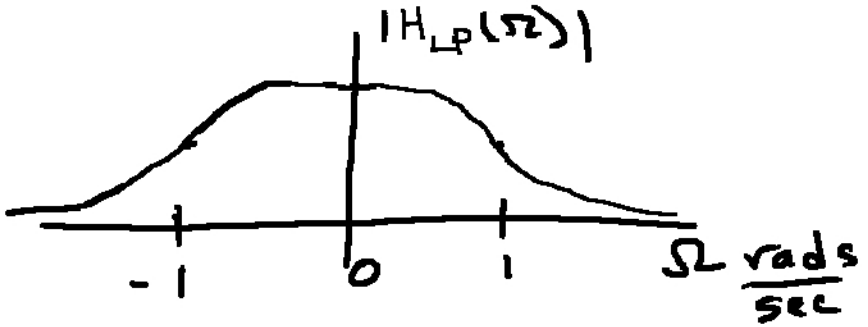
$$\tilde{\Omega}_1 = \frac{-\Omega_0 B + \sqrt{\Omega_0^2 B^2 + 4\Omega_c^2}}{2}$$

$$\tilde{\Omega}_2 = \frac{\Omega_0 B + \sqrt{\Omega_0^2 B^2 + 4\Omega_c^2}}{2}$$

$$\tilde{\Omega}_2 - \tilde{\Omega}_1 = \Omega_0 B$$

B scales width

Example: convert $H_{LP}(s) = \frac{1}{s+1}$ to bandpass $H(s)$ with passband (1/2 power) from $10 \le |\Omega| \le 14$ rads/sec



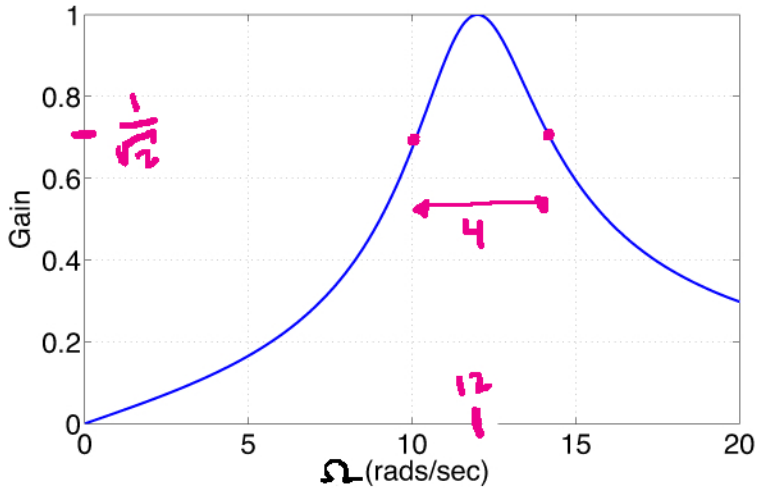
1/2 power bandwidth = 2 rads/sec

Desired 1/2 power bandwidth = 4 rads/sec $\Rightarrow B = 4$

Desired center frequency = 12 rads/sec $\Rightarrow \Omega_c = 12$

$$s = \frac{\tilde{s}^2 + \Omega_c^2}{B\tilde{s}} = \frac{\tilde{s}^2 + 144}{4\tilde{s}}$$

$$H(\tilde{s}) = \frac{1}{s+1}\Big|_{s=\frac{\tilde{s}^2+144}{4\tilde{s}}} = \frac{4\tilde{s}}{\tilde{s}^2 + 4\tilde{s} + 144}$$



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